

HENRY DEUTSCH

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Portfolio: henry-deutsch.com

WORK EXPERIENCE

NewForm AI | Software Engineer I – Full Time

Aug 2025 - Feb 2026

NewForm provides B2B analytics for consumer companies on their ads, synthesized using deep data modeling and AI.

Reported to the CTO and took ownership of 6 major initiatives from inception to implementation in direct collaboration with C-suite

- **Led** major refactor **enabling \$50K+/mo in TikTok ad spend** by incorporating TikTok support for our entire suite of software. Refactored **31K+ LOC in 365 files**, reworking 140+ **tRPC** endpoints & **Bayesian inference** engine for platform-agnostic objects
- Conceived and led ad similarity detection service, authoring white paper and deploying **p-hashing** to **359K ads across 20 clients**
- Built and migrated **100% of internal account managers and external clients** to a streamlined messaging and reporting system

KnoWhiz | Software Engineering Intern

May 2024 - Sep 2024

Developed full-stack features for LLM-based AI learning platform that serves 4.3K+ monthly active users

- Rearchitected full-stack Quizlet import feature to cut large import failures from **~50% to <4%** on 4.3K MAUs, **saving ~200 user hours /mo**. Used **WebSocket**-based progress updates, **Java Spring Boot** batching and two-phase commit patterns
- Led **SEO** improvements and created a gateway Explore page, **increasing monthly organic traffic by ~10%**
- **Mentored** developer in React and TypeScript, cutting PR review cycles by **~50%** and enabling delivery of 12+ production features

HF Engineering | Software Engineering Intern

May 2024 - Sep 2024

Led development of internal onboarding platform enabling 18,000 volunteers to collaborate on open source projects

- Used **AWS Amplify** to set up **CI/CD** pipelines, **reducing design-to-dev handoff time by an avg. of ~30% across 8 teams**
- Created full-stack dashboard for project management with **React (Next.js)** and **JavaScript**, with user authentication

Johns Hopkins University | Software Development Intern

May 2023 - Aug 2023

Built full-stack disaster relief analytics platform for the National Institute of Standards and Technology (NIST)

- Developed a **Python** and **SQL** backend and **Plotly Dash** frontend. Followed **agile sprint** cycles and **Scrum** within a team
- Led database redesign and SQL schema optimization to support new media type uploads, allowing new GPS analysis features
- Integrated **transformer**-based **deep neural network** to analyze audio. Achieved 92% accuracy in extreme-condition classification

EDUCATION

Johns Hopkins University | Bachelor of Science in Computer Science | Baltimore, MD

Aug 2021 - May 2025

Graduated with Departmental Honors in Computer Science

Courses: Full-Stack JavaScript, Artificial Intelligence, Machine Learning, Practical Generative AI, Computer Graphics,

Intermediate Programming (C++), Computer System Fundamentals, Data Structures, Algorithms, Calculus III, Physics II

Awards: Placed 3rd of 43 teams, won \$250 prize at HopHacks Hackathon for [reimagining grocery checkout with computer vision](#)

SOFTWARE PROJECTS

[Chief of NYC Fire Dept Website](#) | Paid Contract — Full Stack Developer

Sep 2021 - Present

- Improved **SEO**, increasing traffic from **<100 to >3.6k visits per mo.** and book royalties from **8K to ~24K YoY (2.99x)**
- Used custom **Java**-based PDF parser to extract structured content from 200+ pages and generate **HTML** and **CSS** layouts
- Developed learning platform with **React** and **JavaScript** with quiz functionality, content management, and progress tracking

[C++ Ray Tracing Engine](#)

- Architected ray tracing engine with **C++**, **OpenGL**, and **multithreading** — achieved 2.5x speedup over single-thread approach
- Engineered 3D rendering pipeline with **GLSL** shaders and parallel **computational geometry** intersection algorithms
- Developed Phong illumination with quaternion camera controls, bilinear **texture mapping**, and ray-traced shadow systems

[N-Body Orbit Simulations](#) | Research at Johns Hopkins

Developed AI-powered orbit analysis system using autoencoder **neural network**, and latent space analysis with clustering algorithms

- Built convolutional autoencoder with **PyTorch** and **NumPy**. Achieved 200:1 compression with .02 MSE loss on orbit data
- Classified orbit paths by their latent space in the NN with **K-means**, **Hierarchical**, **Agglomerative** clustering methods in **Python**
- Deployed scalable ML infrastructure on **AWS EC2** cloud stores enabling parallel processing of 10k+ orbit simulations

TECHNICAL SKILLS

Full-Stack Web Dev: Next.js, React, Node.js, Express.js, Prisma, Redis, tRPC, MongoDB (MERN stack), SQL, Tailwind, Vite

Languages: TypeScript, JavaScript, Python, Java, C++, SQL, HTML, CSS

Other Software: Firebase, Supabase, Git, Jira, Unix/Linux, Kubernetes, Docker, Plotly Dash, AWS (EC2, Amplify), PyTorch